Seminar Proposals

Optimizing Elliptic Curve Arithmetic

- Elliptic Curves and their importance to Cryptography (Definition of a group $E(\mathbb{F}_p)$, Discrete Log and Diffie-Hellman assumptions). See "A Graduate Course in Applied Cryptography", Section 15, https://crypto.stanford.edu/~dabo/cryptobook/BonehShoup_0_6.pdf.
- Projective Elliptic Curve equation (in homogeneous and Jacobian forms), addition law in projective space, why it optimizes classical addition law in affine space. See "Complete addition formulas for prime order elliptic curves", https://eprint.iacr.org/2015/1060.
- Scalar multiplication using double-and-add and non-adjacent form algorithms. Multiple point multiplication (MSM) and its application to polynomial commitment schemes (KZG).
- Optimizing scalar multiplication using efficiently computable endomorphisms. GLV Scalar Multiplication. See "Guide to Elliptic Curve Cryptography", Section 3.5.

Fast Fourier Transform over Finite Fields

- A short remark on structure of $\mathbb{F}_q^\times$ and primitive N-th roots of unity in $\mathbb{F}_q$.
- Lagrange and Barycentric Interpolation formulas. Using N-th roots of unity subgroup for reducing complexity assymptotic.
- Fast Fourier Transform over FFT-friendly finite fields.
- Elliptic Curve Fast Fourier Transform (ECFFT). See "ECFFT: Fast Polynomial Algorithms over all Finite Fields" https://arxiv.org/pdf/2107.08473.

Note: The last point of the lecture requires most preparation. The first three are already presented in my book. One might see a simplified presentation on these topics here: https://zkdl-camp.github.io/files/slides/13-ntt.pdf.

Introduction to Zero-Knowledge Proofs

- What are zero-knowledge proofs and their applications. Interactive Probabilistic Proofs (IP). Soundness, Argument of Knowledge, Zero-Knowledge properties. Fiat-Shamir Heuristic.
- Σ-protocols as an example of classical IP.
- A glimpse into pairing-based zero-knowledge succinct non-interactive argument of knowledge. Schwartz-Zippel Lemma, KZG polynomial commitment scheme. High-level overview of Groth16: algebraic arithmetization and quadratic arithmetic programs.

Note: To avoid using pairings without knowing their algebraic nature, instead of the last point, we might consider zk-SNARKs based on multilinear extensions (Sum-Check and GKR protocols). See https://zkdl-camp.github.io/files/slides/15-sumcheck.pdf.